Raja Roy

Backend Software Engineer | Python · Go · TypeScript | APIs · Distributed Systems · System Design
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SUMMARY

Backend engineer with 4 years of experience building APIs, lending platforms, and compliance systems across fintech and security. Designs stateful, asynchronous workflows with a focus on database design, integrations, and production reliability.

TECHNICAL SKILLS

Languages & frameworks: Python, Go, TypeScript, JavaScript, SQL, FastAPI, NestJS, Node.js, Flask, Django

Databases & messaging: PostgreSQL, MySQL, Redis, RabbitMQ, BullMQ, Elasticsearch

System Design: HLD/LLD, REST APIs, SQL/database design, asynchronous processing, background jobs, caching, idempotency; concurrency and distributed locking concepts

Cloud & tools: AWS EC2, AWS Glue, Docker, Nginx, GitHub, GitHub Actions, RPA Framework, Selenium, Pandas

Familiarity: Temporal (workflow orchestration); Kubernetes (basic hands-on); Prometheus & Grafana (basic)

PROFESSIONAL EXPERIENCE

Razorpay Mar 2025 – Present

Software Engineer — Payroll & Compliance Backend

- Owned PF workflow reliability improvements for 1,500+ organizations, resolving calculation, integration, and recovery failures to help improve automated payment success to ~99%.
- Owned tax-compliance workflow reliability end-to-end; designed Master Transaction Record (MTR) state tracking to associate transactions with payment workflows in a Temporal-orchestrated service. Implemented error classification, retries, reconciliation, and failure recovery for stuck or partial workflows, avoiding duplicate or conflicting processing.
- Developed Quicko-integrated TDS APIs for CSI file handling, FVU generation, filing orchestration, audit logging, idempotent processing, and pagination; migrated Forms 24Q/26Q to 138/140 for revised reporting requirements.
- Automated KAPT calculation and payments for approximately 1,000 organizations, plus ESIC registration and payments for organizations with approximately 50–100 employees each.
- Built a reconciliation engine to match bank payment data against internal ledger records and reduce manual finance operations work.
- Reworked employee onboarding into a draft-first workflow supporting progressive completion, reducing downstream data corrections and support needs.
- Contributed to HLD/LLD, API and schema design, production incident response, deployments, and Prometheus-based alerting.

Kinara Capital Dec 2023 – Mar 2025

Software Engineer — Backend & Lending Platform

- Built a Business Rules Engine for Khatabook and other lending partners handling approximately 1,000–2,000 eligibility requests/day, replacing a paid third-party solution. Combined configurable rules, JWT authentication, DTO validation, Redis caching, database persistence, stateful processing, and service integrations.
- Integrated CKYC search/download through certificate-based external APIs and CRIF credit assessment; implemented existing-customer eligibility checks using customer and loan history.
- Optimized lending APIs through Redis caching, request batching, and SQL optimization; developed asynchronous lending workflows with Python, NestJS, FastAPI, Flask, BullMQ, RabbitMQ, and RPA Framework to reduce manual operations.
- Designed an asynchronous bulk CSV loan-booking approach and service-level architecture targeting approximately 100,000 records/day.
- Received the Q1 FY 2024–25 Certificate of Appreciation for contributions to technology and innovation.

Netsurion Oct 2022 – Nov 2023

Associate Python Developer — Security Engineering

- Built a Python-based security detection tool analyzing Elasticsearch event streams; implemented real-time alerts for port/host scans, attack patterns, and login anomalies across Linux and Windows to support proactive threat investigation.

SELECTED PROJECT

RoleOS · AI Job Search Platform · Personal Project · Live: roleos-eight.vercel.app

Built a FastAPI/PostgreSQL job-search platform with resume parsing, staged discovery, deterministic matching and a LangGraph Career Copilot. Added multi-user isolation, JWT/Google SSO, email OTP, approved outreach and admin analytics. Deployed on GCP Cloud Run, Cloud SQL and GCS with a React/TypeScript frontend.

EDUCATION

B.E., Electronics & Communication · Technocrats Institute of Technology, Bhopal · 2016 – 2020 · CGPA: 7.12